RGB-D Rail Surface Defect Inspection Driven by Conditional Diffusion Architecture and Frequency Knowledge

条件扩散 频率知识

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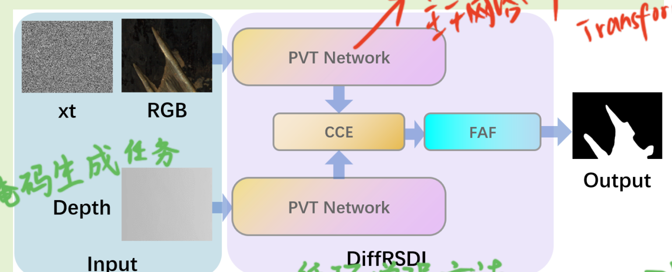
***Abstract*—RGB-D Rail Surface Defect Inspection (RSDI) is a critical measure for ensuring transportation safety. It improves inspection accuracy by using depth maps, but ① the issue of poor-quality depth maps in rail defect datasets is often overlooked. Additionally, ② the similarity between the foreground and background has always been a persistent challenge in RGB-D RSDI. In this article, we propose a novel network, termed DiffRSDI to overcome these challenges in RGB-D RSDI, which treats RGB-D RSDI as a mask generation task (掩码生成任务) using a conditional diffusion architecture. Specifically, DiffRSDI consists of two key components: Cross-modal Cyclic Enhancement (CCE) module and Frequency-aware Alternate Fusion (FAF) module. CCE module is based on the cyclic enhancement approach (循环增强方法), and aims to eliminate interference (消除干扰) from poor-quality depth maps and aggregate (聚合) cross-modal features. FAF module overcomes the similarity between the foreground and background in the frequency-domain, and achieves cross-scale fusion (跨尺度融合) in the fused features. Comprehensive experiments conducted on the NEU RSDDS-AUG dataset confirm that our method achieves superior performance in comparison to 14 related methods. The DiffRSDI code and results are available at https://github.com/zeroyi37/DiffRSDI.**



主干网络(Pyramid vision Transform)

***Index Terms*—Rail surface defect inspection, conditional diffusion architecture, frequency knowledge, RGB-D salient object detection.**

I. Introduction

RAIL transportation has significantly facilitated the travel and daily lives of people in various countries. However, as the speed of high-speed trains increases, the stress on railway tracks also intensifies (加剧). The quality of railway tracks is crucial for both transportation safety and travel security, necessitating regular inspections by railway management to detect surface defects promptly (迅速地). Common defects on tracks include scar defects and hole defects, and timely detection of these defects is crucial to prevent railway transportation accidents, ensuring safety and minimizing economic losses.

Traditional rail surface defect inspection (RSDI) typically requires experienced railway staff to manually inspect each section of track, consuming considerable human resources and lengthy work periods. Due to the significant differences in the shape and size of rail surface defects, as well as the high similarity between the foreground and background on the rail surface, the difficulty of traditional RSDI methods has greatly increased, making traditional manual inspection methods insensitive and inefficient.

Significant progress has been achieved in rail surface defect inspection in recent years [1]–[6]. However, these methods often struggle to accurately detect genuine (真正的) defects when the rail has similar foreground and background characteristics. To address this issue, depth maps have been introduced by researchers as supplementary information, where defects on the depth maps of tracks differ from the normal surface, enabling better differentiation between the foreground and background. This has led to the considerable attention on RGB-D RSDI, with the development of numerous related methods [7]–[11]. However, there are some unsatisfactory aspects in current RGB-D RSDI methods: (1) a portion of

This work was supported in part by the National Natural Science Foundation of China under Grant 62401350 and Grant 62471285, in part by the Shanghai Sailing Program under Grant 24YF2713000, and in part by the Open Research Fund of Anhui Province Key Laboratory of Machine Vision Detection and Perception under Grant KLMVI-2024-HIT-04.

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the depth maps are of poor-quality, which can negatively impact inspection accuracy, and 2) The similarity between the foreground and background, makes it difficult to detect defects in regions with similar backgrounds.

Given these considerations, in this paper, we attempt to introduce the diffusion architecture [12] into RGB-D RSDI by treating the RGB-D RSDI task as a generative task and propose a novel framework named DiffRSDI, leveraging 利用 the powerful generative capability of diffusion architecture to address these challenges. The conditional information provided to the diffusion architecture is crucial for its performance. Therefore, we propose the Cross-modal Cyclic Enhancement (CCE) module, which aggregates 聚合 depth features and RGB features, then enhances them through a cyclic enhancement approach to overcome the issue of poor-quality depth maps. Additionally, we propose the 频率感知型交替融合 Frequency-aware Alternate Fusion (FAF) module, which effectively achieves cross-scale fusion and introduces additional complementary information through 频域处理 frequency-domain processing. This approach allows the proposed DiffRSDI to tackle poor-quality depth maps, utilize depth information as a complementary modality effectively, and perform well despite foreground-background similarity.

创新点

Our main contributions are summarized as follows:

- We apply a conditional diffusion architecture to form predictions for RGB-D RSDI for the first time, which successfully highlights specific rail surface defects.
- We propose a Cross-modal Cyclic Enhancement (CCE) module to effectively aggregate and enhance cross-modal features through the cyclic enhancement approach, and overcome the interference from poor-quality depth maps.
- We propose a Frequency-aware Alternate (FAF) Fusion module to integrate more useful frequency-domain information and effectively achieve cross-scale fusion.

II. Related Work

A. Rail Surface Defect Inspection

Rail surface defect inspection is a critical task in ensuring the safety and reliability of railway transportation. In the field of rail surface defect inspection, various methodologies have been proposed to evaluate rail quality in time. Recently, researchers have adopted many methods to inspect rail surface defect [13]–[16]. Zhang *et al.* [4] introduced a segmentation network that leverages multiple context information to inspect no-service rail surface defect. Yang *et al.* [17] developed an end-to-end, multilevel approach for rapid detection of rail surface defects. Liu *et al.* [18] presented a convolutional neural network based on pyramid features designed for rail surface defect detection. Li *et al.* [19] presented an innovative segmentation network for detecting no-service rail surface defects, utilizing normalized attention and dual-scale interaction. Xu *et al.* [20] presented a model leveraging self-supervised representation learning for RSDI, utilizing a convolutional encoder–decoder neural network to accurately detect rail defects. Guo *et al.* [21] presented an enhanced framework that utilizes a lightweight backbone and attention mechanism for the automatic detection of rail surface defects.

The RGB-D RSDI task focuses on detecting rail surface defects by utilizing both depth map and RGB image. In recent years, this task has garnered significant attention, leading to the emergence of many promising methods [7]–[11]. Wu *et al.* [7] presented a network based on depth repeated-enhancement to extract spatial data from depth maps and detailed features from RGB images for surface defect inspection of rails. Wang *et al.* [8] introduced a knowledge distillation-based progressive enhancement network for efficient detection of rail surface defects. Huang *et al.* [10] proposed an efficient skeleton-aware network for detecting surface defects at the pixel level. Zhou *et al.* [11] presented a knowledge distillation-based modal evaluation network to improve the feature representation for rail defect detection. Zhou *et al.* [22] implemented a hierarchical exploration approach in their RSDI network. Wang *et al.* [23] presented a collaborative learning attention-based multimodality network to perform rail defect inspection and built a RGB-D dataset NEU RSDDS-AUG because of the scarcity of defective data. Wang *et al.* [24] presented a Depth-assisted semi-supervised network to alleviate the annotation burden for RGB-D RSDI. Yan *et al.* [25] proposed a network that integrates autocorrelation and specificity for rail defect detection, aiming to explore both the autocorrelation characteristics within individual modalities and their specificities.

However, these methods have not addressed the issue of poor-quality depth maps in RGB-D RSDI datasets. Some RGB-D SOD methods [26]–[30] have begun to focus on the problem of poor-quality depth maps. Inspired by these observations, we propose a cyclic enhancement approach. By leveraging the cyclic enhancement-based CCE module in cross-modal feature interaction, we reduce the negative impact of poor-quality depth maps and improve RGB-D RSDI accuracy.

B. RGB-D Salient Object Detection

显著目标检测

RGB-D salient object detection (SOD) focuses on identifying and segmenting visually distinct objects within an image using both RGB and depth information. Traditional RGB-D SOD methods [31]–[33] have recently shown competitive performance. Ju *et al.* [34] presented a method that leverages anisotropic center-surround differences for depth map analysis. Cong *et al.* [26] presented a method to assess depth map reliability and applied it to mitigate the impact of poor depth maps. Recent advancements have seen the adoption of deep learning techniques [35]–[42]. Wang *et al.* [43] developed two-streamed convolutional neural networks, with each stream responsible for feature extraction and saliency map prediction via a saliency fusion module. Zhai *et al.* [44] proposed a bifurcated backbone strategy network, which uses bifurcated backbone strategy and depth-enhanced method for SOD. Bhuyan *et al.* [45] presented an RGB-D SOD method designed to capture multilevel features by incorporating diverse contexts as well as long-range dependencies.

However, unlike the goal of RGB-D SOD, which detects and highlights the most salient objects in the visual input, RGB-D RSDI seeks to inspect defect regions that are highly similar to the background. Simply applying the context aggregation

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strategies from RGB-D SOD to fuse different levels of features is not suitable for real-world RGB-D RSDI and fails to address the issue of foreground-background similarity. In this paper, we propose a novel feature fusion module named FAF for RGB-D RSDI from a frequency-domain perspective, combining frequency and spatial domains to overcome the negative impact of foreground-background similarity.

C. Diffusion Models 扩散模型

Recently, diffusion models have gained significant attention as a powerful class of generative models. Unlike traditional models, the diffusion architectures based on Markov chain, and denoise data samples by learning the denoising process. Diffusion models have demonstrated great performance in image synthesis [12], [46]. Rombach *et al.* [46] used diffusion architectures as powerful generators for high-resolution synthesis in a convolutional manner. Whang *et al.* [47] presented a network utilizing conditional diffusion architectures to address image deblurring. Despite those achievements, diffusion architectures also have great performance in image segmentation, edge detection and camouflaged object detection. Wu *et al.* [48], [49] proposed the first diffusion probabilistic model toward image segmentation applications in general medical. Chen *et al.* [50] proposed the first diffusion architecture for general edge detection, which applies diffusion probabilistic model in the latent space and enables the classic cross-entropy loss. Ye *et al.* [51] used the diffusion architecture's denoising process to progressively refine the mask for camouflaged object detection. Zhou *et al.* [52] proposed a surface defect detection framework based on diffusion architecture, combining a diffusion probabilistic model and a pre-trained backbone. Wang *et al.* [53] proposed a new semi-supervised learning approach for visual measurement, built on a diffusion probabilistic model.

However, there are no studies that explore the effectiveness of diffusion architecture in the RGB-D RSDI task. In this paper, we propose a denoising diffusion-based network DiffRSDI for RGB-D RSDI task.

III. Proposed Model

A. Method Overview

Fig. 1 illustrates the pipeline of our proposed denoising 去噪 diffusion-based DiffRSDI, which consists of three components: 1) Feature encoding, 2) Cross-modal Cyclic Enhancement (CCE) module, and 3) Frequency-aware Alternate Fusion (FAF) module.

In the first component, the Pyramid Vision Transformer (PVT) [54] is utilized as the backbone to perform feature encoding in our approach. As illustrated in Fig. 1, the network takes an RGB image F, a depth map D and a noise mask 噪声掩膜 x_t as input. To align dimensions 调整尺寸 with the RGB image, the depth map is replicated across three channels. The encoder is composed of two symmetrical PVT networks, with the RGB image and depth map processed independently through PVT structure. We introduce the noise mask xt in the RGB branch of the PVT. Each stage's output features from the RGB and depth branches are denoted as F_R^i and F_D^i, $(i = 1, 2, 3, 4)$, respectively. These feature maps are processed through a CCE module to purify 净化 depth features, generating enhanced RGB features F_{RD}^i. Then an FAF module processes the enhanced features to fuse the cross-scale feature maps. The denoising network, featuring a simple U-shaped structure, takes the fused features as conditional features to produce the denoised mask prediction F_{out}. The input RGB image is set to $352 \times 352 \times 3$, while the depth map is configured as $352 \times 352 \times 1$, respectively.

B. Preliminaries

In this work, we treat the RGB-D RSDI tasks as a mask generation paradigm 掩码生成范式 and approach RGB-D RSDI from a noise-to-image perspective. We formulate it as a diffusion architecture and employ a conditional diffusion network to generate predictions.

Our DiffRSDI is based on the diffusion architecture, which includes two essential processes: the forward process and the reverse process. During the forward process, the noise is progressively introduced to the data over time steps t, where t runs from 1 to T. Given the training data x_0, the forward process produces a sequence 一系列 of noisy samples $\{x_t\}_{t=1}^T$ by iteratively 迭代地 adding Gaussian noise:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I), \tag{1}$$

where $\beta_t \in (0, 1)$ represents a variance schedule 方差时间表 that dictates the noise level introduced at each time step. The closed-form representation of the forward process is given as:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I), \tag{2}$$

where $\bar{\alpha}_t = \Pi_{s=1}^t \alpha_t$, $\alpha_t = 1 - \beta_t$.

The reverse process is modeled to undo the diffusion process, gradually removing the noise and reconstructing the original data sample from x_T, which is approximately Gaussian noise. The reverse process is parameterized as:

$$p(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; u_\theta(x_t, t), \sigma_t^2 I), \tag{3}$$

where $\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$, and the mean $u_\theta(x_t, t)$ is learned by a neural network and is typically parameterized as:

$$u_\theta(x_t, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\hat{x}_0, \tag{4}$$

where $\hat{x}_0$ is the result of our model.

C. Cross-modal Cyclic Enhancement Module

Depth maps contain crucial complementary information for RGB-D RSDI, which plays a crucial role in distinguishing genuine features. However, distance cues 距离线索 in the poor-quality depth maps may fail to provide reliable information or even introduce incorrect interference. To address the issue of poor-quality depth maps, we propose the Cross-modal Cyclic Enhancement (CCE) module based on a cyclic enhancement approach. The CCE module is an essential component of the DiffRSDI network, primarily responsible for aggregating features from both the depth and RGB branches and cyclically enhancing these features. It contains two cyclic enhancement parts, where each cyclic enhancement first strengthens the depth features using the RGB features, and then enhances

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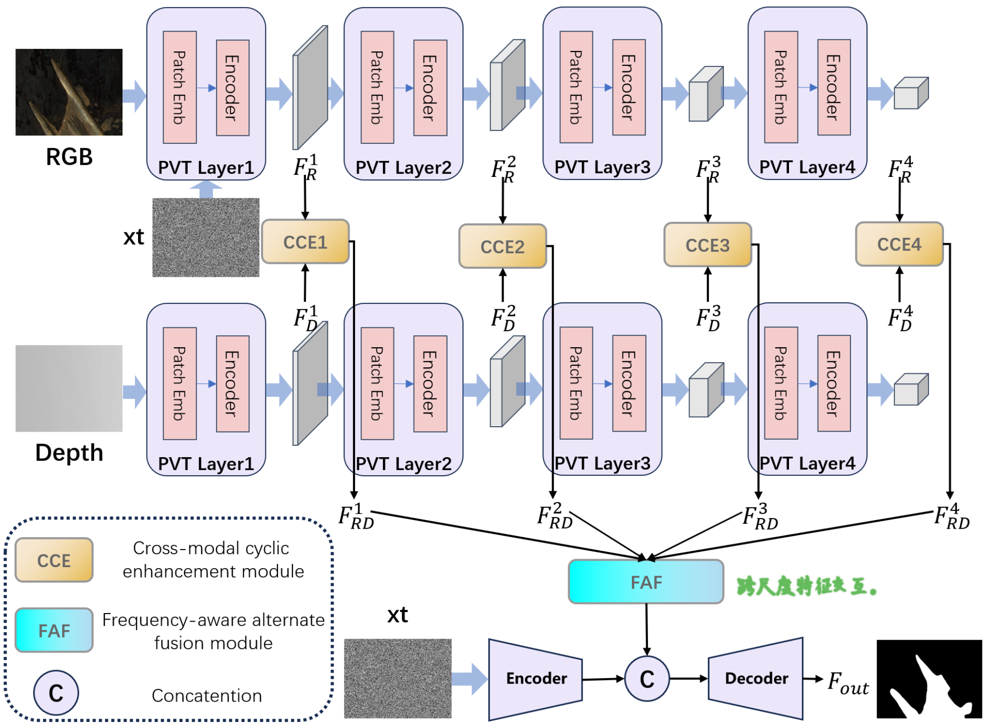

Fig. 1. Pipeline of the proposed DiffRSDI. First, the input RGB image, depth map and noise mask x_t are extracted by the Pyramid Vision Transformer to obtain cross-modal features. At each stage, the cross-modal features from the RGB and depth branches are processed through a Cross-modal Cyclic Enhancement module to cyclically enhance cross-modal features. Then a Frequency-aware Alternate Fusion module processes these enhanced features, extracting low-frequency components from cross-scale features and effectively fusing them to generate conditional features. Finally, the input noise mask x_t and the conditional features are fed to an encoder and decoder to generate results.

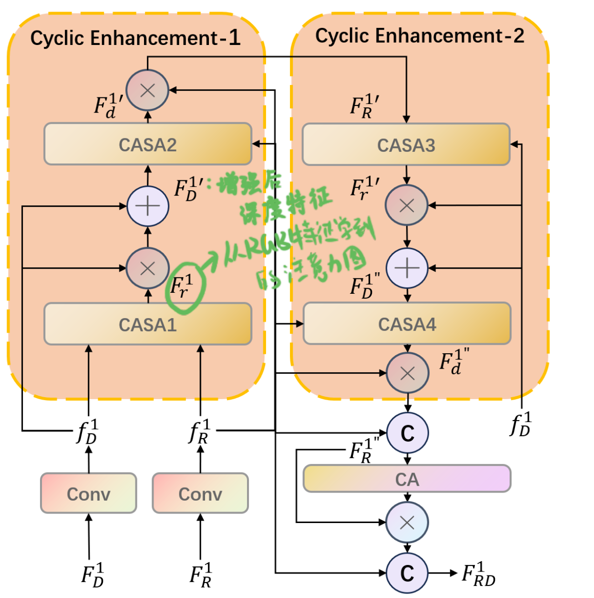

Fig. 2. Illustration of Cross-modal Cyclic Enhancement (CCE) module.

the RGB features using the enhanced depth features. Through the CCE module, we obtain features that undergo two cyclic enhancements from depth to RGB, and then from depth to RGB, thereby reducing the negative impact of poor-quality depth maps. The details of CCE1 are illustrated in Fig. 2.

In CCE module, we employed several attention mechanisms to enhance these features, including spatial attention (SA) [55] 空间注意 and channel attention (CA) [56] operations. As shown in Fig. 3, we aggregate CA operation and SA operation within the CASA module to obtain more effective feature attention. We further interact the features of both modalities after applying channel attention operation. The features from both branches are first multiplied and then added to the original features. Subsequently, the interacted features are multiplied with the input features of both the RGB and depth branches, resulting in more deeply integrated features. The process of the CA part in CASA module can be described as:

$$\begin{cases} F_{CR}^i = \{[CA(f_R^i) \otimes CA(f_D^i)] \oplus CA(f_R^i)\} \otimes f_R^i \\ F_{CD}^i = \{[CA(f_R^i) \otimes CA(f_D^i)] \oplus CA(f_R^i)\} \otimes f_D^i. \end{cases} \tag{5}$$

For spatial attention, we apply a similar interaction method as used in channel attention. The process of the SA part in CASA module can be described as follows:

$$\begin{cases} F_r^i = Conv_7\{Cat[SA(F_{CR}^i) \otimes SA(F_{CD}^i), SA(F_{CR}^i)]\} \\ F_d^i = Conv_7\{Cat[SA(F_{CR}^i) \otimes SA(F_{CD}^i), SA(F_{CD}^i)]\}, \end{cases} \tag{6}$$

where $Cat(\cdot)$ represents the cross-channel concatenation operation, $Conv_7(\cdot)$ represents the 7×7 convolution operation, $\otimes$ is the element-wise multiplication, and $\oplus$ is the element-wise summation.

Each cyclic enhancement process in our approach includes two CASA modules. Specifically, we first apply the CASA module to the RGB features and depth features in the Cyclic

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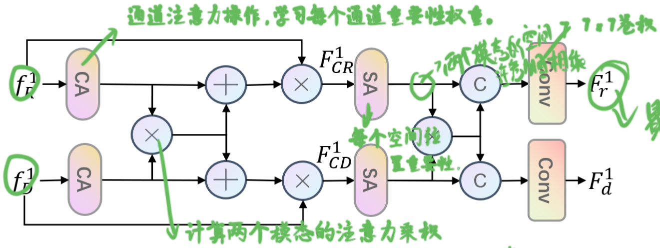

Fig. 3. Illustration of CASA module.

再分别与对应的注意力相加
然后与原始特征相乘得到增强后的特征

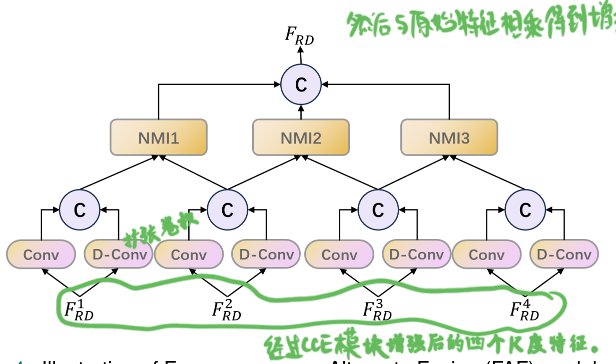

Fig. 4. Illustration of Frequency-aware Alternate Fusion (FAF) module.

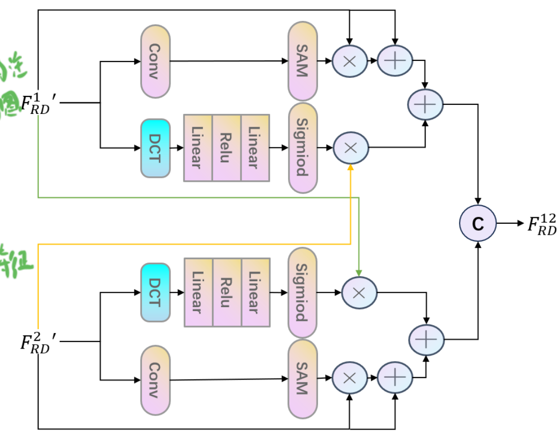

Fig. 5. Illustration of Neighborhood Multi-scale Interaction (NMI) module.

Enhancement-1, achieving SA map F_r^i from RGB features, which is used to modulate depth features for reducing interference factors. This process can be computed as:

$$\begin{cases} F_r^i = CASA(f_R^i, f_D^i) \\ F_D^{i'} = F_r^i \otimes f_D^i \oplus f_D^i, \end{cases} \quad (7)$$

where F_r^i represents the output of the first CASA module.

Then we add modulated depth features to the CASA module to obtain SA maps $F_d^{i'}$ from depth features, these SA maps are used to enhance RGB features, which can be computed as:

$$\begin{cases} F_d^{i'} = CASA(f_R^i, F_D^{i'}) \\ F_R^{i'} = F_d^{i'} \otimes f_R^i, \end{cases} \quad (8)$$

where $F_d^{i'}$ represents the output of the second CASA module.

We have already enhanced the RGB features once, but instead of directly feeding the enhanced features into the subsequent network, we apply a similar enhancement process again to obtain more representative features. This process is referred to as Cyclic Enhancement-2, which can be described as:

$$\begin{cases} F_D^{i''} = CASA(F_R^{i'}) \otimes f_D^i \oplus f_D^i \\ F_R^{i''} = Cat[CASA(F_D^{i''}) \otimes f_R^i, f_R^i], \end{cases} \quad (9)$$

After finishing all cyclic enhancement processing, we apply the CA operation to achieve a more valuable feature F_{RD}^i. Then we stack f_R^i on the features after CA operation to preserve the original RGB contents. This operation can be formulated as:

$$F_{RD}^i = Cat[F_R^{i''} \otimes CA(F_R^{i''}), f_R^i]. \quad (10)$$

D. Frequency-aware Alternate Fusion Module

In RGB-D RSDI, the similarity between the foreground and background has long been a challenging issue. Inspired by [57]–[59], we observe that processing feature maps from a frequency-domain perspective helps resolve the issue of foreground-background similarity. Therefore, we propose the Frequency-aware Alternate Fusion (FAF) module to address this issue, which uses discrete cosine transform (DCT) to integrate more useful frequency-domain information and introduce additional complementary information from the frequency-domain, thereby uncovering clues that help distinguish the foreground from the background. Then, the frequency-domain branch is combined with spatial attention, enabling the model to focus not only on the important frequency components in the frequency-domain but also on critical parts in the spatial domain.

离散余弦变换

As shown in Fig. 4, the proposed FAF module contains several convolutions and two submodules called the neighborhood multi-scale interaction (NMI) module. First, we perform convolution and dilated convolution on four features from CCE modules. Then we integrate these features and obtain four input features of NMI modules. As the NMI1 shown in Fig. 5, the feature maps of two adjacent passes through both spatial and frequency branches, the domain features processed by the DCT transform are used to interact with the features of the other branch, enhancing the effectiveness of low-frequency information in the feature map. After the feature maps pairwise pass through the NMI module, we perform cross-channel concatenation operation to integrate feature maps from different NMI modules.

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相邻传递

The frequency branch operation in the NMI module can be formulated as:

$$F_f^i = F_{RD}^{j'} \otimes \varphi(Lin(ReLu(Lin(DCT(F_{RD}^i))))), \quad (11)$$

where $\varphi(\cdot)$ denotes the sigmoid function, $Lin(\cdot)$ refers to the linear transformation operation, $ReLu(\cdot)$ represents the ReLU activation function, and $DCT(\cdot)$ represents the 2D discrete cosine transform.

In the spatial branch, we employ the SA module to capture detailed position information. These components from the spatial and frequency branches are combined by the element-wise addition operation, then we integrate these neighborhood

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TABLE I
QUANTITATIVE PERFORMANCE COMPARISON OF OUR METHOD AND OTHER 14 STATE-OF-THE-ART METHODS ON NEU RSDDS-AUG DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN.

Models	NEU RSDDS-AUG								
	Sm	MAE	adpEm	meanEm	maxEm	adpFm	meanFm	maxFm	WeiFm
ACSD [34]	0.5560	0.3605	0.4989	0.4941	0.6698	0.4567	0.4675	0.5739	0.3804
BBSNet [44]	0.8273	0.0733	0.8394	0.9043	0.9090	0.8607	0.8560	0.8667	0.8405
CDCP [60]	0.5721	0.2361	0.6810	0.5885	0.6926	0.5806	0.5090	0.5901	0.4352
CLANet [23]	0.8338	0.0686	0.8633	0.9122	0.9207	0.8622	0.8632	0.8771	0.8455
CoNet [36]	0.7854	0.1008	0.8615	0.8654	0.8795	0.8296	0.8288	0.8336	0.8004
DCMC [26]	0.4828	0.2872	0.6419	0.4708	0.5931	0.4831	0.3555	0.4958	0.3224
DF [35]	0.5636	0.2408	0.7041	0.5102	0.7125	0.6288	0.4399	0.6357	0.3963
DMRA [61]	0.7343	0.1404	0.8229	0.8149	0.8343	0.7645	0.7603	0.7825	0.7136
FHENet [22]	0.8376	0.0634	0.8664	0.9212	0.9261	0.8702	0.8689	0.8818	0.8560
HAINet [27]	0.7171	0.1710	0.7143	0.7812	0.8278	0.7899	0.7585	0.8051	0.7410
S^2MA [43]	0.7752	0.1411	0.7519	0.7890	0.8635	0.6890	0.7433	0.8168	0.6899
DRERNet [7]	0.8455	0.0588	0.8685	0.9289	0.9330	0.8794	0.8783	0.8908	0.8679
SAINet [25]	0.8514	0.0554	0.8747	0.9333	0.9363	0.8906	0.8898	0.8975	0.8799
CamoDiffusion [51]	0.8576	0.0565	0.8556	0.9312	0.9361	0.8826	0.8805	0.8944	0.8774
DiffRSDI	0.8674	0.0485	0.8721	0.9426	0.9457	0.9020	0.9000	0.9095	0.8936

features after combining and obtain the output neighborhood aggregation feature.

E. Loss Function

二元交叉熵损失

We compute the weighted binary cross-entropy (BCE) loss and weighted intersection-over-union (IoU) loss on the prediction and the ground truth to form our final loss function:

$$\mathcal{L}_{total} = \mathcal{L}^{w}_{BCE} + \mathcal{L}^{w}_{IoU}, \tag{12}$$

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where $\mathcal{L}^{w}_{BCE}$ represents the weighted BCE loss, $\mathcal{L}^{w}_{IoU}$ represents the weighted IoU loss.

IV. EXPERIMENTS

A. Implementation Details

We implement our proposed DiffRSDI by PyTorch using two NVIDIA GeForce 3090 GPUs. The input image is resized to $352\times352\times3$ to accommodate the input requirements of the model. Training continues with a batch size of 12 for 150 epochs. To adjust the learning, we set the initial learning rate to 10^{-4}. For testing, a conditional diffusion architecture is used to iteratively denoise samples generated from a standard normal distribution across T steps, where our default configuration used T=10 for the sampling process. We do not adopt any other post-processing methods.

B. Datasets and Evaluation Metrics

In our experiments, we use a no-service rail RGB-D dataset, NEU RSDDS-AUG dataset [23]. The proposed DiffRSDI is trained and tested on the NEU RSDDS-AUG dataset. The datasets contain 1862 images, divided roughly in an 8:2 ratio, with 1500 images used for training and 362 images for testing.

We use nine metrics for quantitative evaluation, including S-measure (Sm) [62], mean absolute error (MAE), maximum E-measure (maxEm), adaptive E-measure (adpEm), mean E-measure (meanEm), maximum F-measure (maxFm) [63], adaptive F-measure (adpEm), mean F-measure (meanEm), and weighted F-measure (WeiFm) [64].

C. Comparison with State-of-the-arts

We compare our method with 14 state-of-the-art methods, including ACSD [34], BBSNet [44], CDCP [60], CLANet [23], CoNet [36], DCMC [26], DF [35], DMRA [61], FHENet [22], HAINet [27], S^2MA [43], DRERNet [7], SAINet [25], and CamoDiffusion [51]. To ensure a fair comparison, the saliency maps for all methods are either supplied by the authors or generated by running their publicly available codes.

定量比较

1) Quantitative Comparison: For the 14 state-of-the-art methods, we report the quantitative performance comparison of our method and them on NEU RSDDS-AUG dataset in terms of nine metrics. The quantitative comparisons are presented in Table I. Our proposed method surpasses most previous state-of-the-art methods on NEU RSDDS-AUG dataset, further validating the effectiveness of our proposed DiffRSDI.

2) Qualitative Comparison: We represent the visual comparison results of our proposed method with state-of-the-art representative models in Fig. 6. In particular, the rail background of the first five rows is very similar to its surrounding, while the rail background is complex in the last three rows. It can be observed that when the rail background is complex and similar to the foreground, our DiffRSDI still achieves accurate RGB-D RSDI results. This is due to the FAF module, which ensures comprehensive cross-scale information fusion, significantly reducing the negative impact of the similarity between the foreground and background on RGB-D RSDI.

Moreover, all the depth maps in these visual comparison results are of poor-quality. Even when they fail to provide valuable distance cues or even offer incorrect information, DiffRSDI is still capable of precise defect localization, with

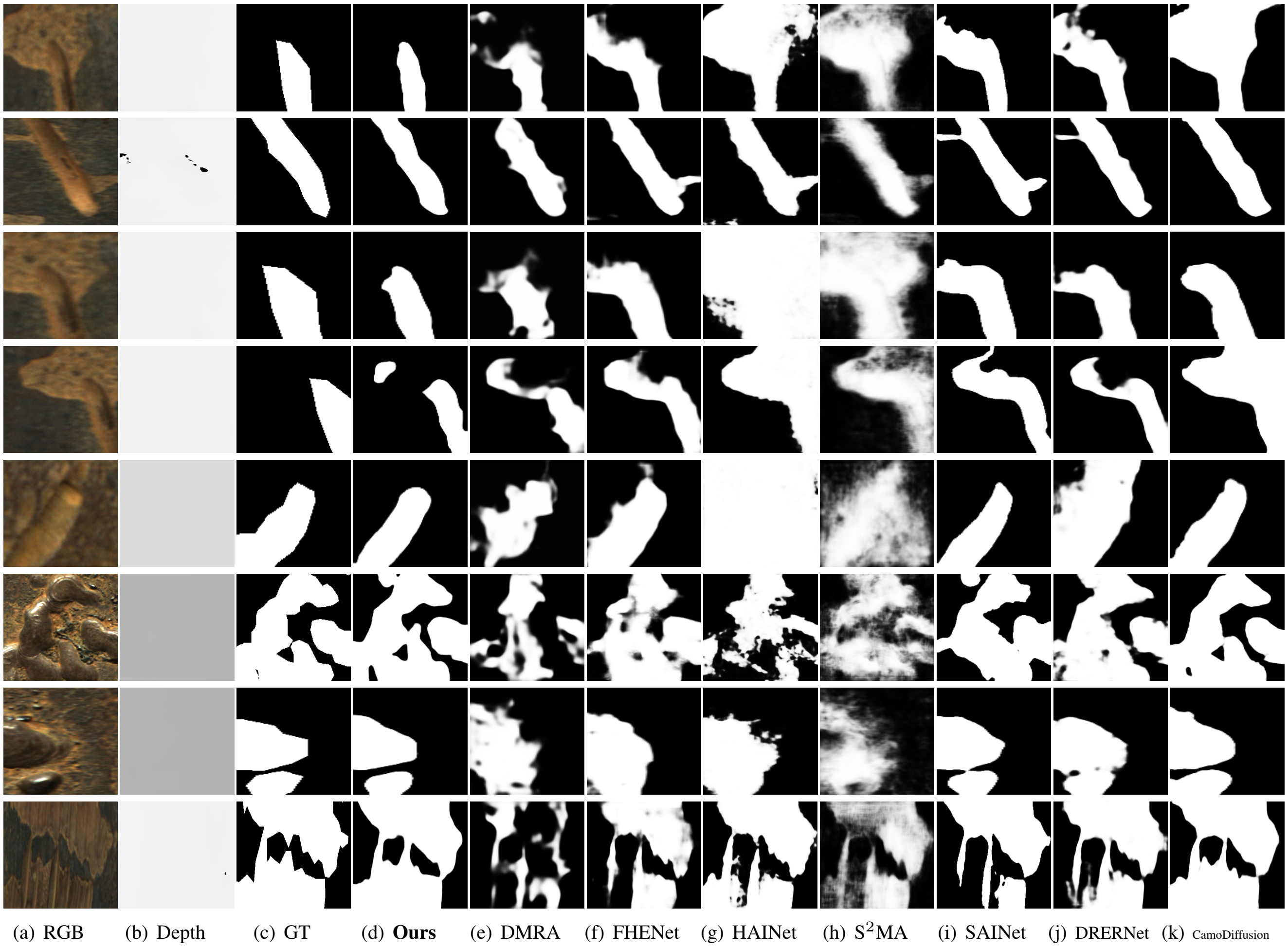

Fig. 6. Qualitative Comparisons of RGB-D RSDI with seven last methods, including DMRA, FHENet, HAINet, S^{2}MA, SAINet, DRERNet, CamoDiffusion.

complete areas in the detected defects. This is attributed to our CCE module, which effectively enhances RGB information using depth information, fully leveraging the complementary role of depth information in RGB-D RSDI.

D. Ablation Studies

This subsection provides comprehensive ablation studies on the NEU RSDDS-AUG dataset to evaluate the contribution of each critical component in our approach. Specifically, we investigate 1) the impact of different modules, 2) the importance of CCE, 3) the effectiveness of the cyclic enhancement process in CCE, 4) the importance of FAF, and 5) the rationality of the structure of frequency and spatial branches in FAF.

1) The Impact of Different Modules: In this subsection, we perform extensive ablation experiments to verify the significance of the various modules in the proposed DiffRSDI. Table II presents the results of the module evaluation. The baseline model (*i.e.,* base) includes PVT network and a basic diffusion model. We can observe that using the CCE module results in a significant improvement compared to the standard baseline model. It indicates that CCE module is generally an effective strategy for RGB-D RSDI. FAF module, which is built upon CCE module, introduces additional complementary information from the frequency-domain, thereby further improving the detection performance.

TABLE II
THE IMPACT OF DIFFERENT MODULES.

Modules			NEU RSDDS-AUG			
base	CCE	FAF	Sm	MAE	maxEm	maxFm
✓			0.8543	0.0539	0.9367	0.8956
✓	✓		0.8642	0.0510	0.9420	0.9037
✓	✓	✓	**0.8674**	**0.0485**	**0.9457**	**0.9095**

2) The Importance of CCE: The CCE module is essential for both fusing RGB and depth features and enhancing the RGB features. To study its importance, we replaced the CCE module with convolution and element-wise summation operations(*i.e.,* CCE-SUM) and replaced the CCE module with convolution and concatenation operations (*i.e.,* CCE-CAT). The results of the ablation studies are provided in Table III, the obvious decline in the performance of CCE-SUM and CCE-CAT demonstrates that our CCE module outperforms direct operations (*i.e.,* CCE-SUM and CCE-CAT) for the fusion of

TABLE III
ABLATION ANALYSES FOR THE IMPORTANCE OF CCE AND THE EFFECTIVENESS OF THE CYCLIC ENHANCEMENT PROCESS IN CCE.

Models	NEU RSDDS-AUG			
	Sm	MAE	maxEm	maxFm
CCE-CAT	0.8630	0.0506	0.9436	0.9009
CCE-SUM	0.8606	0.0510	0.9424	0.9057
CE-0	0.8609	0.0523	0.9399	0.9025
CE-1	0.8630	0.0507	0.9423	0.9060
CE-3	0.8597	0.0524	0.9399	0.9009
Ours (CE-2)	**0.8674**	**0.0485**	**0.9457**	**0.9095**

TABLE IV
ABLATION ANALYSES FOR THE IMPORTANCE OF FAF AND THE RATIONALITY OF THE FREQUENCY AND SPATIAL BRANCHES STRUCTURE IN FAF.

Models	NEU RSDDS-AUG			
	Sm	MAE	maxEm	maxFm
FAF-CAT	0.8642	0.0510	0.9420	0.9037
FAF-SUM	0.8639	0.0505	0.9429	0.9049
FAF-S	0.8632	0.0519	0.9410	0.9051
FAF-F	0.8602	0.0518	0.9409	0.9027
Ours	**0.8674**	**0.0485**	**0.9457**	**0.9095**

RGB and depth features, underscoring the critical importance of the CCE module.

3) The Effectiveness of the Cyclic Enhancement Process in CCE: To analyze the effectiveness of the cyclic enhancement process, we replaced the complete cyclic enhancement process in CCE module without cyclic enhancement process (*i.e.,* CE-0), only one cyclic enhancement process (*i.e.,* CE-1) and multiple cyclic enhancement process (*i.e.,* CE-3). The multiple cyclic enhancement process consists of three cyclic enhancement processes. As shown in Table III, the performance of the CCE module gradually improves with up to two cyclic enhancement processes. However, when multiple cyclic enhancement processes are applied, the performance of CE-3 is degraded compared with other cyclic enhancement methods. This indicates that using two cyclic enhancement processes is the most effective, as RGB features produced by our CCE module are of higher quality. This confirms the effectiveness of the complete cyclic enhancement process in CCE module.

4) The Importance of FAF: We conducted an ablation study to analyze the importance of FAF during the feature fusion process. To validate the importance of FAF, we replaced the FAF module with element-wise summation and convolution operations (*i.e.,* FAF-SUM) and replaced the FAF module with concatenation and convolution operations (*i.e.,* FAF-CAT). Table IV presents the results of the ablation studies. Through comparison with the performances of FAF-SUM and FAF-CAT, the performance of FAF module shows an obvious improvement. We can discover that the FAF module is highly effective in the fusion of features across different scales.

5) The Rationality of The Structure of Frequency and Spatial Branches in FAF: To explore the necessity of the structure of frequency and spatial branches in FAF, we conducted experiments where the frequency branch was replaced with convolution operations (*i.e.,* FAF-S) and the spatial branch was replaced with convolution operations (*i.e.,* FAF-F). As reported in Table IV, the performances of FAF-S and FAF-F are similar, but neither match the performance of the FAF module. This demonstrates that using both branches together is the most effective method, which confirms the rationality of the structure of frequency and spatial branches in FAF.

E. Failure Cases and Analyses

We show some failure cases of our DiffRSDI and conduct an analysis. As shown in Fig. 7, there are two challenging scenarios of our DiffRSDI: 1) adjacent continuous defect scenarios (the frist two rows) and 2) multiple similar defect scenarios (the last two rows). In the first two rows, there are multiple adjacent continuous defects. Our method tends to misjudge the boundaries of these adjacent defects, failing to accurately highlight all defects and fully detect each defect's boundary, leading to missing boundary details. In the last two rows, the image contains multiple similar defects, but our method potentially missing some defects. These undetected defects are less prominent compared to others in the image, causing our method to mistakenly classify them as part of the rail background, resulting in incomplete defect detection.

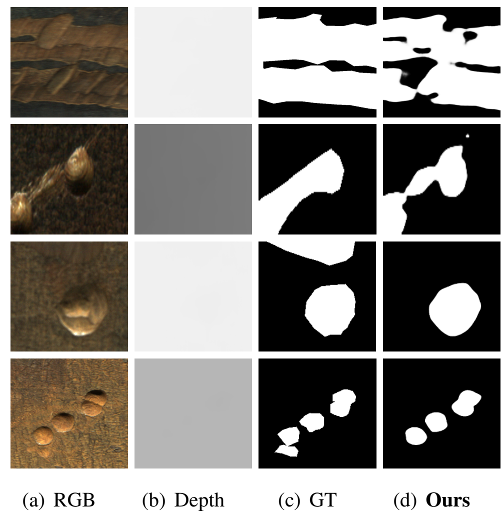

Fig. 7. Some failure cases of our DiffRSDI. The frist two rows represent scenarios with adjacent continuous defects, while the last two rows show scenarios with multiple similar defects

V. CONCLUSION

In this paper, we present a novel diffusion-based network, named DiffRSDI, for RGB-D rail surface defect inspection. DiffRSDI is composed of Cross-modal Cyclic Enhancement (CCE) module and Frequency-aware Alternate Fusion (FAF) module, and utilizes the denoising diffusion-based framework to generate saliency maps. In particular, CCE module is designed to extract the complementary information from different modalities and mitigate the negative effects of poor-quality depth maps through a cyclic enhancement approach. FAF module is designed to fuse the cross-scale feature maps effectively and overcome the negative impact of foreground-background similarity. Experimental results indicate that our model performs competitively compared with 14 state-of-the-art methods.

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